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26 September 2026

The Head

National Centre for Medium Range Weather Forecasting

Ministry of Earth Sciences

A-50, Sector 62, Noida, Uttar Pradesh 201309

Subject: Request for access to NEPS ensemble output for research on Smart India Hackathon 2026 problem statement SIH26079, "AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts"

Respected Sir/Madam,

I am writing on behalf of a student team at Chandigarh University working on Smart India Hackathon 2026 problem statement SIH26079, which was contributed by NCMRWF. The team has built a working prototype and would like to request access to NEPS output so that the method can be demonstrated on India's own operational ensemble.

The starting point for the work was a measurement made against India's own gauge observations. Ensemble spread — the quantity a duty forecaster uses today to judge confidence — appears to explain about 44% of the variance in forecast error when all forecasts are pooled together, but under 1% once forecasts in the same rainfall regime are compared with one another. The pooled figure largely reflects the fact that spread grows with rainfall amount, not that it identifies which forecasts will fail.

Our system learns that judgement instead. Because NEPS reforecast archives are not publicly available, it was developed and validated on the NOAA GEFSv12 reforecast, verified against IMD 0.25° gauge-gridded daily rainfall over 1,252 gauge-observed gridpoints. Across 104 forecast dates spanning 2018 and 2019, it outperforms ensemble spread by +0.149 in stratified ROC-AUC, with the entire 95% confidence interval above zero, and it is ahead in every rainfall band tested. The same model scores a live operational forecast cycle, unmodified, in under two minutes end to end.

The method is ensemble-agnostic by construction: it consumes ensemble mean and spread fields and does not depend on the model that produced them. Our request is therefore narrow — access to NEPS output so that the result can be reproduced on the ensemble NCMRWF actually operates, rather than on an American one. We would be grateful for whichever of the following is feasible, listed in descending order of what it would ask of your team:

1. NEPS hindcast or archived forecast fields over a multi-year window;
2. approximately 50 to 100 archived NEPS forecast dates, spread across seasons;
3. ensemble mean and spread only, with no individual members, for those dates; or
4. guidance on the correct procedure by which such a request should be made.

The fields required are listed in the annexure overleaf. The data would be used solely for this research. We will not redistribute it, we are glad to sign any undertaking NCMRWF requires, and we will share our

code, results and verification with the Centre. We would also be glad to present the work should that be of interest.

I may add that these addresses were taken from NCMRWF's published RTI disclosure, as no general research contact is listed on the Centre's website. I apologise if this is not the correct channel, and would be grateful if the letter could be directed to the appropriate officer.

Thank you for your time and consideration.

Yours sincerely,

Pramatma Vishwakarma

[Designation], Department of [Department]

Chandigarh University

On behalf of Team Hellfire Labs: Devansh, Ankit, Paras, Arnav, Mahika and Dhruv.

Enclosures: Annexure A — data specification (overleaf). Technical note — method and evaluation protocol, two pages, sent with this letter. A prototype dashboard is available at vishwas-five.vercel.app

Annexure A

Data specification

Problem statement SIH26079 · Team Hellfire Labs, Chandigarh University

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The model consumes ensemble mean and ensemble spread derived from the fields below. Individual members are not required if mean and spread can be supplied directly.

Field	Level	Used for
Geopotential height	500 hPa	Upper-air pattern and its gradient
u-wind	500 hPa	Mid-level flow; 500–850 hPa shear
v-wind	500 hPa	Mid-level flow; flow deformation
u-wind	850 hPa	Low-level monsoon flow; shear
v-wind	850 hPa	Moisture transport; shear
Precipitable water	Entire atmosphere	Atmospheric moisture and its anomaly
CAPE	Surface	Instability and its anomaly
Accumulated precipitation	Surface	Forecast rainfall amount and anomaly

Sampling requirements

- Lead times: daily steps from Day 1 to Day 10.
- Domain: the Indian land region, approximately 5–40°N and 65–100°E.
- Horizontal resolution: whatever is convenient. The present work uses 0.5°, and the method does not depend on the grid.
- Initialisation: a single cycle per date is sufficient; 00 UTC preferred for consistency with the existing work.
- Format: GRIB2 or NetCDF, either is equally usable.

Verification data

None is required. Rainfall verification uses IMD 0.25° gauge-gridded daily rainfall, which the team already holds through the public historical archive.

Undertaking

The data will be used solely for the research described in the covering letter. It will not be redistributed, published as raw fields, or used for any commercial purpose. The team will comply with any data policy, licence or undertaking NCMRWF requires, and will share its code, results and verification with the Centre.

Contact for this request: Pramatma Vishwakarma, Chandigarh University · [official email] · [phone]